

Advance modding

Advance modding involves the creation of new advances - the Europa Universalis V representation of technological progress.

Please help with verifying or updating older sections of this article.
At least some were last verified for version 1.0.

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Technical details

Advance definitions are stored in common/advances. They are usually placed in the common top order.



Syntax

Most advances are comprised of the following syntax:

```

ostentatious_clothing = {                # Key of the advance
    icon = fla_cloth
    age = age_5_absolutism

    requires = lead_ore_dressing

    global_alum_output_modifier = 0.1
    global_dyes_output_modifier = 0.1
}

```

Age requirement

Every advance requires the age argument to determine which age tab the advance belongs to.

Modifiers

See also: [Modifier modding](#), [Modifier types](#)

Advances are implicit modifier blocks, which allow modifier types that affect the nation once the advance has been researched.

An advance can also be given modifier types that are applied during the research process though - using `modifier_while_progressing` blocks. There can be a number of these blocks which can hold any number of modifier types. Each block has a `scale` value, a `script` value that determines the strength of the modifier, and `potential_trigger`, which determines if the modifier applies at all. E.g:

```

modifier_while_progressing = {
    scale = {
        value = 1          # 100% of the modifier is given at a time
        add = {            # which can be increased by another 100% based on legitimacy
            value = legitimacy
            divide = 100
        }
    }
    potential_trigger = {   # but only if the country is a monarchy
        government_type = government_type:monarchy
    }
    monthly_prestige = 0.05
}

```

Other requirements

Advances can have only a single prerequisite advance. This creates a simple tree structure with a number of "root" advances that branch out to a number of "children". Prerequisite advances are defined using `requires` and must be defined before the advance requiring it. Advances without a defined prerequisite are semi-randomly assigned to the tree. To ensure that a given advance never becomes a "parent" `allow_children = no` can be used. This prevents randomly assigned child advances and throws an error if another advance lists it as a prerequisite.



potential to limit the advance to that geographical region while most "root" advances use allow to require the related institution to be embraced.

While an advance can "graphically" require only one advance, other advances can be required in the allow trigger block.

Besides that, there are special flags for certain types of requirements, namely government and country type. These are referenced in the content tab as well as highlighted with an icon.

Token name	Right-side argument	Right-side argument source	GUI script function	Additional information
government	Government Type	common/government_types	AdvanceItem.IsGovernmentUnlock	Requires the country to have the selected government type for potential and shows up under the "Government Type" bonuses in main menu content tab
country_type	Country Type	location, building, pop, army, navy	AdvanceItem.IsCountryTypeUnlock	Requires the country to be of the country type for potential and shows up under the "Country Type" bonuses in main menu content tab

Additionally, the game implicitly assigns advances to additional categories based on triggers in potential:

Category	Trigger	GUI script function	Other information
"Country" advance	Presence of tag or has_or_had_tag triggers in potential.	AdvanceItem.IsCountryUnlock	
"Religion" advance	Presence of religion comparison triggers in the potential	AdvanceItem.IsReligionUnlock	

Note for using culture_group requirement: The Content tab in country selection only shows advancements belonging to alphabetically first culture group the country's culture belongs into.

Controlling tree generation



ance trees are generated and cannot be structured on their own. There are, however, several factors to influence how the tree gets generated, though.



be loaded before the advance that requires them.

Controlling AI bias

There are two ways to control what the AI will research - `ai_weight` and `ai_preference_tags`.

`ai_weight` is a scripted value evaluated on the country which adds additional weight for the country AI to research this advance specifically.

`ai_preference_tags` is a set of preference tags that are combined with country's setup research preferences to increase the AI's likelihood to research this advance by that amount.

Research cost

Every advance has its research cost. The base research cost is determined by the BASE_RESEARCH_COST define (Vanilla value: 25), but can be further modified using `research_cost` which accepts a floating point value, which then gets added on top of the original cost - `research_cost = 1.0` will give a penalty of +100.0%. Moreover, there is also a hardcoded research cost increase based on ages. If an advance for an outdated age is being researched, a reduction of PREVIOUS_AGE_REDUCTION define (Vanilla value: -8) is applied. Additionally, every age ahead of traditions gets a penalty of AGE_RESEARCH_MODIFIER define (Vanilla value: 0.15) times the amount of ages that passed.

Unlocking capabilities

Advances provide an easy interface for unlocking many different scripted types. These do not require any allow and potential checks and the tooltip for the advance is automatically updated to mention the types:





unlock_employment_system	Employment system	common/employment_systems	
unlock_unit	Unit type	common/unit_types	
unlock_ability	Unit ability	common/unit_abilities	
unlock_interaction	Character interaction	common/character_interactions	
unlock_country_interaction	Country interaction	common/country_interactions	
unlock_estate_privilege	Estate privilege	common/estate_privileges	
unlock_relation_type	Relation type	common/scripted_relations	
unlock_building	Building type	common/building_types	
unlock_law	Law	common/laws	
unlock_policy	Policy	common/laws	
unlock_levy	levy type	common/levies	
unlock_heir_selection	Heir selection	common/heir_selections	
unlock_government_reform	Government reform	common/government_reforms	
unlock_casus_belli	Casus belli	common/casus_belli	
unlock_cabinet_action	Cabinet action	common/cabinet_actions	
unlock_subject_type	Subject type	common/subject_types	
unlock_production_method	Production method	common/building_types	
unlock_diplomacy	Diplomatic action	Not extendable	Uses different syntax - lists. This is also the case if there is only one diplomatic action to be unlocked.

Starting technology level

Main article: [Setup modding § Starting technology level](#)

`starting_technology_level` is an integer that is used to setup starting research for countries.

Icon



This section requires verification emperical tests to determine how the unit and levy icons are assigned(Or link to a wiki page which describes it properly).. Please add appropriate references.

See the [Icon priority](#) section on the talk page for details about what needs verification.

The game looks for an icon in the following order:

1. first `unlock_unit` line (some part of the illustration of the unit which seems to be based on the category and the `gfx_tags`)
2. first `unlock_levy` line (probably uses a similar algorithm as `unlock_unit`)
3. building icon of the first `unlock_building` line



It was the named after the advance

Icon attribute

icon attribute is used to set a custom icon key for the advance if none of the unlock_ attributes exist which give an icon.

Advance key as filename

Otherwise, the advance key is used as the icon name.

Icon path

Advance icons are stored in ADVANCE_ICON_PATH define (Vanilla value: "gfx/interface/advance").

In order to have an icon assigned, a new .dds entry entitled with the key must be added there.

Example

For example:

```
zan_gold_trade_of_africa = {  
    ...  
}
```

```
coffea_arabica = {  
    icon = origin_of_coffee  
    ...  
}
```

The first advance uses zan_gold_trade_of_africa.dds as its icon, while the latter uses origin_of_coffee.dds.

Ages event and new unlocks

for attribute is used to set an advance as unlocked by a specific chosen age preference. It, therefore, accepts one of three values:

- adm
- dip
- mil

Age-preference advances get a special texture in the GUI, which is done with AdvanceItem.IsChoiceUnlock.

Setting age preference

preference is set using set_age_preference effect, which sets the age preference for the ent age.



See also: [Localization](#)

Advances only need two keys for localization:

- `<key>`
- `<key>_desc`

Script involving advances

- [research_advance](#) effect in country scope researches the advance scope on the right-hand side:
- [advance_type](#): is a global data [scope link](#) used to fetch the advance type for a given key.
- [can_research_advance](#) is a country scope trigger that checks if the country can research a not-yet researched advance scope on the right.
- [has_advance](#) is a country scope trigger that checks if the country has the advance identified by the key.
- [has_advance_available](#) is a country scope trigger that checks if the country has the advance identified by the key researched or it is available to research.
- [num_of_advances_researched](#) is a country scope value trigger that checks if the country has researched a certain amount of advances.
- [has_advance_for_employment_system](#) is a country scope value trigger that checks if the country has researched an advance that has a [unlock_employment_system](#) that * points towards towards the scope on the right that evalutes to an [employment_system](#).
- [has_advance_for_succession_law](#) is a country scope value trigger that checks if the country has researched an advance that has a [unlock_heir_selection](#) that points towards towards the scope on the right that evalutes to an [heir_selection](#).

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"https://eu5.paradoxwikis.com/index.php?title=Advance_modding&oldid=30344" sayfasından alınmıştır

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